Gene Pulser® Electroprotocol

Cell Type Bacterial, gram negative

Species Used *E. coli*, LE 392, DH5α

Molecules Electroporated DNA: mostly yeast shuttle vectors, Bluescript™ -type vectors

---

Before the Pulse

Cell Growth Medium LB Medium

Growth Phase at Harvest O.D. (600) = 0.5

Pre-pulse Incubation 10 to 20 sec. on ice

Wash Solution Distilled deionized water. Cells frozen in 10% glycerol

The Pulse

Instruments Used Gene Pulser® apparatus

Electroporation Temperature Ice, 0 °C

Electroporation Medium 10 % glycerol

Cell Density 10 (10) cells / ml

Volume of Cells 40 µl

DNA Concentration 1 to 100 ng

DNA Resuspension Buffer SOC

Volume of DNA 0.5 to 1 µl

Cuvette Gap 0.2 cm

Voltage 2.5 kV

Field Strength 12.5 kV/cm

Capacitor 25 µF

Resistor 200 Ω (Pulse Controller)

Time Constant 4.5 to 4.7 msec

After the Pulse

Outgrowth Medium SOC

Outgrowth Temperature 37 °C

Length of Incubation 30 to 40 mins.

Selection Method or Assay Used Ampicillin resistance

Electroporation Efficiency 10 (7) to 10 (10) transformants/µg DNA

Per Cent Survival Not given

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

SOC: 2% Bacto tryptone, 0.5% Bacto yeast extract, 10mM NaCl, 2.5mM KCl, 10 mM $MgCl_2$, 10 mM $MgSO_4$, 20 mM glucose.

LB: 1% Bacto tryptone, 0.5% Bacto yeast extract, 0.5% NaCl.

Name of Submittor Dr. D. Sanglard

Institution Address Swiss Federal Institute of Technology
Inst. for Biotechnologie
ETH - Hoenggerberg
8093 Zurich
Switzerland

Survey Number 049